Question ID: c7a187a7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

$$f(x) = x^2 - 18x - 360$$

If the given function f is graphed in the *xy*-plane, where $y = f(x)$, what is an *x*-intercept of the graph?

Answer

A. $(-12, 0)$

B. $(-30, 0)$

C. $(-360, 0)$

D. $(12, 0)$

Correct Answer: A

Rationale

Choice A is correct. It's given that $y = f(x)$. The *x*-intercepts of a graph in the *xy*-plane are the points where $y = 0$. Thus, for an *x*-intercept of the graph of function f, $0 = f(x)$. Substituting 0 for $f(x)$ in the equation $f(x) = x^2 - 18x - 360$ yields $0 = x^2 - 18x - 360$. Factoring the right-hand side of this equation yields $0 = (x + 12)(x - 30)$. By the zero product property, $x + 12 = 0$ and $x - 30 = 0$. Subtracting 12 from both sides of the equation $x + 12 = 0$ yields $x = -12$. Adding 30 to both sides of the equation $x - 30 = 0$ yields $x = 30$. Therefore, the *x*-intercepts of the graph of $y = f(x)$ are $(-12, 0)$ and $(30, 0)$. Of these two *x*-intercepts, only $(-12, 0)$ is given as a choice.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.